

# Decision and Risk Analysis

## Lecture 1: **Bayesian Thinking for Risk and Decision Analysis**

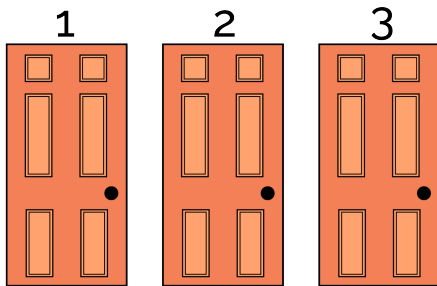
DAEN 427/ISEN 427/ISEN 627

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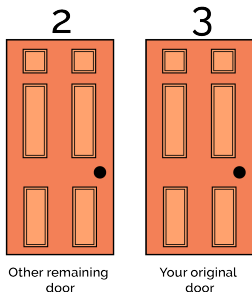


There are **three doors** for you to choose between:  
**two** contained **nothing** of value and **one** contained a **prize**.



You **pick one** of three closed doors,  
making your guess as which door contained the prize.

Then, a door you **did not** choose is revealed,  
and it **does not** contain the prize.



With two doors still closed (including the one you originally picked),  
you are given the choice:

- do you want to **swap doors**,
- or **stay** with the one you picked?

Your Decision: Should you stay or switch?

A decision problem begins with a **choice to be made** under certain **objectives** and **constraints**.

## Risk and Uncertainty

- **Choice under risk:** Outcome probabilities are known (or can be estimated reliably).
- **Choice under uncertainty:** Outcome probabilities are unknown (or difficult to estimate).

## Decision and Risk Analysis

1. **Modeling Decisions:** Define the available choices or actions.
2. **Modeling Risk:** Analyze situations where outcome probabilities are known.
3. **Modeling Uncertainty:** Analyze situations where outcome probabilities are unknown (or difficult to estimate).
4. **Modeling Preferences:** Represent how decision makers value outcomes and trade-offs.

## Decision and Risk Analysis, Examples

### 1. **Modeling Decisions**

A student chooses between studying for an exam tonight or attending a social event.

### 2. **Modeling Risk**

A company launches a new product knowing there is a 70% chance of high demand and a 30% chance of low demand.

### 3. **Modeling Uncertainty**

A farmer decides what crop to plant without knowing how much rain will fall this season.

### 4. **Modeling Preferences**

A traveler chooses between a cheaper flight with a long layover and a more expensive direct flight.

# Review

## Exploratory data analysis

1. Exploratory data analysis, or EDA, is an initial data analysis that summarizes main characteristics. It often is done through visual means or in basic summary statistics.
2. Data visualization can be done through many ways, and helps us see patterns and identify potential issues or interesting patterns.



## Categorical data

### **Nominal data**

1. Named categories with no inherent order or numeric meaning.
2. When there are only two categories, the variable is often called binary or dichotomous
3. Examples: Material type (steel, aluminum, concrete), Model outcome (approved, rejected), Class label (spam, not spam)

### **Ordinal data**

1. Categories with a natural order, but where the spacing between levels is not necessarily measurable or equal.
2. Examples: Risk level (low, medium, high), Damage severity (minor, moderate, severe), Rating (poor, fair, good, excellent)

## Numeric data

### **Count or rank data**

1. (hopefully self-explanatory)
2. Examples: number of defects, number of clicks, number of missing values.

### **Continuous data**

1. Measurable quantities where difference between possible values can be arbitrarily small.
2. Data might lie within a range or be unbounded (in either direction)
3. Examples: temperature, pressure, time spent, latency, model error.

## Some complications

Suppose you're designing a study and want to measure smoking exposure. We might treat it as:

1. Nominal: yes/no
2. Ordinal: current vs. former vs. never smoker
3. Count: number of cigarettes smoked in past week
4. Continuous: calculation of lifetime pack-years smoked

In the real world, decisions are made based on statistical considerations such as modeling interpretations, likelihood of measurement error, or simply convenience and real-world burden of collecting such data.

## The population vs. a sample

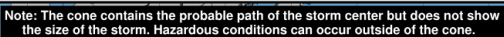
1. A population is the entire group that you want to draw conclusions about.
2. A sample is the specific group that you will collect data from. The size of the sample is always less than the total size of the population.
3. In research, a population doesn't always refer to people. It can mean a group containing elements of anything you want to study, such as objects, events, organizations, countries, etc.

## Parameters

1. Attribute of the population of interest.
2. Not computable directly.  
(unless entire population is perfectly measured)
3. Usually written in Greek letters.

## Statistics

1. Attribute of a sample.
2. Function of the observed values at hand.
3. Usually written in Roman letters.



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## Point estimates hide uncertainty

A forecast location such as

$$\hat{x}_t$$

is only a summary of a full posterior predictive distribution:

$$p(X_t \mid D).$$

The probability distribution of the storm's location  $X_t$  at time  $t$ , given the observed data  $D$ .

As the forecast horizon increases, uncertainty about the storm's future location typically increases as well.

## Location uncertainty is not the same as risk

The quantity relevant for risk analysis is broader than the probability distribution of the storm center:

$$p(\text{wind, rain, surge, flooding at location } s \mid D).$$

Risk can therefore remain substantial even when a location is outside the most probable storm track.



## Bayesian decisions combine probability and consequence

For an action  $a$ , evaluate posterior expected loss:

$$\mathbb{E}[L(a, \theta) \mid D] = \int L(a, \theta) p(\theta \mid D) d\theta.$$

A low-probability outcome may still justify precaution if its consequences are sufficiently severe.

**Point estimate  $\rightarrow$  Posterior uncertainty  $\rightarrow$  Hazard distribution  $\rightarrow$  Risk-informed decision**

## Interpretations of probability



*There is a 1 in 3 chance of selecting a white ball*



*The surgery has a 50% probability of success*

## Interpretations of probability

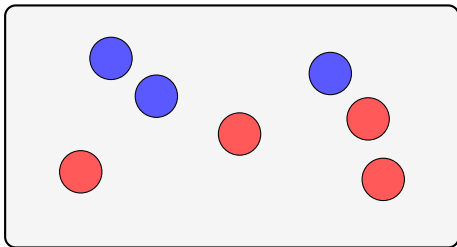
### **Long-run frequency**

- A probability describes what would happen over many repetitions of the same random process.

### **Degree of belief**

- A probability describes uncertainty about a single event, given available information.

## A simple random experiment



If one ball is selected at random, what is the probability it is **red**?

**Probability spaces** are objects that model random experiments - real-world processes involving states that occur "randomly".

A probability space consists of three components:

1. A **sample space**, containing all possible outcomes.
2. Subsets of the sample space, called **events**, which comprise any number of possible outcomes (including none of them!).
3. A function that assigns **probabilities** to events.

An event is said to **occur** if the outcome of the random experiment is contained in that event.

## Sample space

Depends on what we observe.

| Random experiment            | Sample space                       |
|------------------------------|------------------------------------|
| Toss one coin                | $\{H, T\}$                         |
| Toss two coins               | $\{HH, HT, TH, TT\}$               |
| Roll two dice and sum        | $\{2, 3, 4, \dots, 12\}$           |
| Time until component failure | $[0, \infty)$                      |
| Classify an email            | $\{\text{spam}, \text{not spam}\}$ |

[https://www.reddit.com/r/learnmath/comments/qp7ulb/what\\_does\\_0\\_infinity\\_mean\\_is\\_it\\_r\\_numbers\\_from/](https://www.reddit.com/r/learnmath/comments/qp7ulb/what_does_0_infinity_mean_is_it_r_numbers_from/)

## Events

Are subsets of the sample space.

They are the statements to which we assign probabilities.

| Sample space             | Example event                       |
|--------------------------|-------------------------------------|
| $\{H, T\}$               | The coin lands heads                |
| $\{HH, HT, TH, TT\}$     | At least one head occurs            |
| $\{2, 3, 4, \dots, 12\}$ | The sum is odd or equal to 8        |
| $[0, \infty)$            | Failure occurs after one year       |
| Defect categories        | The part is classified as defective |



## Probability

Is a number between 0 and 1 assigned to an event.

- $P(\text{heads in one fair coin toss}) = 0.5$
- $P(\text{at least one head in two fair coin tosses}) = 0.75$
- $P(\text{sum is 8 when rolling two dice}) = 5/36$
- $P(\text{machine fails before inspection})$  may be estimated from data or a model

## Events as subsets

Remember, events are subsets of the entire sample space. Let's take for now the example of tossing a single fair coin and recording the outcome. For one fair coin toss, the sample space is  $S = \{H, T\}$ .

We can define the simple events  $A$  and  $B$  in the outcome space:

- $A$ : getting a head,  $A = \{H\}$ .
- $B$ : getting a tail,  $B = \{T\}$ .

Other possible events include:  $\emptyset, \{H, T\}$ . *What do the empty event and the full sample space mean in words?*

## Set operations for events

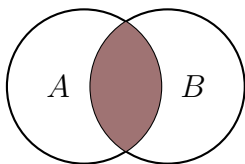
For two events  $A$  and  $B$ :

- **Intersection:**  $A \cap B$  means  $A$  and  $B$  both occur.
- **Union:**  $A \cup B$  means  $A$  or  $B$  occurs, including both.
- **Complement:**  $A^c$  means  $A$  does not occur.
- **Set difference:**  $A \setminus B$  means  $A$  occurs but  $B$  does not.

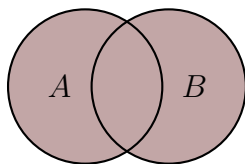
Events  $A$  and  $B$  are **disjoint** or **mutually exclusive** if

$$A \cap B = \emptyset.$$

Intersection:  $A \cap B$



Union:  $A \cup B$



## Kolmogorov axioms

A probability rule  $P(\cdot)$  must satisfy three basic axioms.

1. For every event  $A$ ,  $P(A) \geq 0$ .
2. The probability of the entire sample space is  $P(S) = 1$ .
3. If  $A$  and  $B$  are disjoint, then

$$P(A \cup B) = P(A) + P(B).$$

These axioms are enough to derive many useful probability rules.

[https://en.wikipedia.org/wiki/Probability\\_axioms](https://en.wikipedia.org/wiki/Probability_axioms)

## Two useful rules ...

From the axioms, we obtain two rules used constantly.

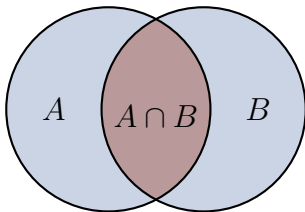
### **Complement rule**

$$P(A^c) = 1 - P(A).$$

### **Inclusion-exclusion**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Why subtract the intersection?



When we add  $P(A) + P(B)$ , the overlap is counted twice.  
Inclusion-exclusion subtracts one copy of the overlap.

## De Morgan's laws

Complements, unions, and intersections can be combined:

### **Complement of a union**

$$(A \cup B)^c = A^c \cap B^c$$

### **Complement of an intersection**

$$(A \cap B)^c = A^c \cup B^c$$

The rules can be expressed in English as:

- The negation of "A or B" is the same as "not A and not B".
- The negation of "A and B" is the same as "not A or not B".



## Example: engineering reliability

A manufacturing line classifies parts using two screening tests.

- $A$ : a part fails the visual inspection.
- $B$ : a part fails the sensor-based inspection.

Suppose

$$P(A) = 0.08, \quad P(B) = 0.12, \quad P(A \cap B) = 0.03.$$

Then the probability that a part fails at least one inspection is

$$P(A \cup B) = 0.08 + 0.12 - 0.03 = 0.17.$$

## Example: data science classification

A model flags observations for review.

- $A$ : the observation is truly anomalous.
- $B$ : the model flags the observation.

Important event statements include:

- $A \cap B$ : true positive.
- $A^c \cap B$ : false positive.
- $A \cap B^c$ : false negative.
- $A^c \cap B^c$ : true negative.



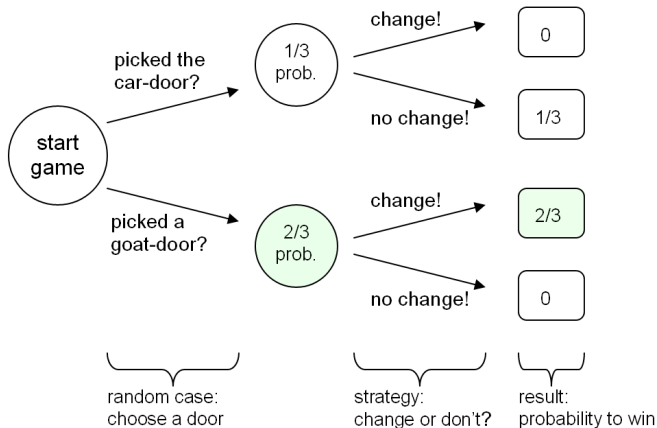
Probability gives a formal language for **uncertainty**.

- A probability space consists of outcomes, events, and probabilities.
- Events are sets, so set operations matter.
- The complement rule and inclusion-exclusion are basic tools for probability calculations.

These ideas support inference, modeling, and risk analysis.

# The Monty Hall Problem

Switching **Doubles** Your Chance of Winning!



Refs.: Wikipedia and University of Illinois  
Code examples